Predicting YouTube Views of Licensed Music Videos Based on Spotify Streams and Song Features

Chengrui Hao, Tiaotiao Jiang, Jing Chang, Yuming Gao

1. Introduction

In today's music industry, music videos are no longer just an accessory to a song — they are an essential driver of branding, popularity, and cultural impact. Think of Michael Jackson's *Thriller* or Beyoncé's *Single Ladies*: the music itself was groundbreaking, but the videos elevated these works into unforgettable cultural moments. Even in recent years, artists such as Lil Nas X with *Old Town Road* or Psy with *Gangnam Style* have shown how a music video can go viral and, in turn, catapult the song into global stardom.

Platforms like YouTube, now the most widely used streaming hub for music videos, have accelerated this trend, allowing visual storytelling to shape how audiences experience music. More often than not, we see the video and the song feed into each other's success — a viral video can push a track to the top of the charts, while an already popular track can generate millions of views when paired with a compelling video.

For artists and record labels, this interplay has made music videos a central piece of marketing campaigns, often determining whether a release gains fleeting attention or lasting influence. This raises an important question: what kinds of songs are most likely to generate a successful, widely viewed music video?

To explore this, our project merges Spotify and YouTube data across the top 10 songs of more than 1,000 artists. By analyzing audio features and streaming patterns, we aim to predict YouTube view counts from Spotify characteristics, shedding light on the traits that make a song most suitable for visual amplification.

2. Data Cleaning

Our dataset, sourced from Kaggle, contains 20,713 rows and 31 columns, each representing a unique song. Alongside basic identifiers such as artist, album, and URL, it provides variables capturing both musical qualities and commercial performance.

Key variables include Spotify audio features and streaming metrics. The features — danceability, energy, key, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and duration (ms) — offer a rounded characterization of a song's sound profile applicable across genres, enabling insights into diverse musical styles and artists.

On the performance side, the dataset covers each song's reach on Spotify (streams) and YouTube (views, likes, comments). This dual-platform view allows us to explore how intrinsic audio characteristics and listening behaviors relate to video engagement, providing a comprehensive lens on cross-platform popularity.

To prepare the dataset, we removed irrelevant columns such as video descriptions and URLs, then deleted rows missing key variables — Url_youtube, Stream, and Danceability. Since each song is unique, interpolation risked bias, making deletion preferable. We retained only licensed, official music videos, excluding unverified or unofficial uploads. Videos containing "lyric" in their titles were also removed, as these represent lyric visualizers rather than full music videos.

After cleaning, we obtained a refined dataset of 13,200 songs — a balanced foundation for exploration and modeling, ensuring our analysis focuses on authentic, comparable videos and strengthening the validity of our insights.

3. Exploratory Data Analysis

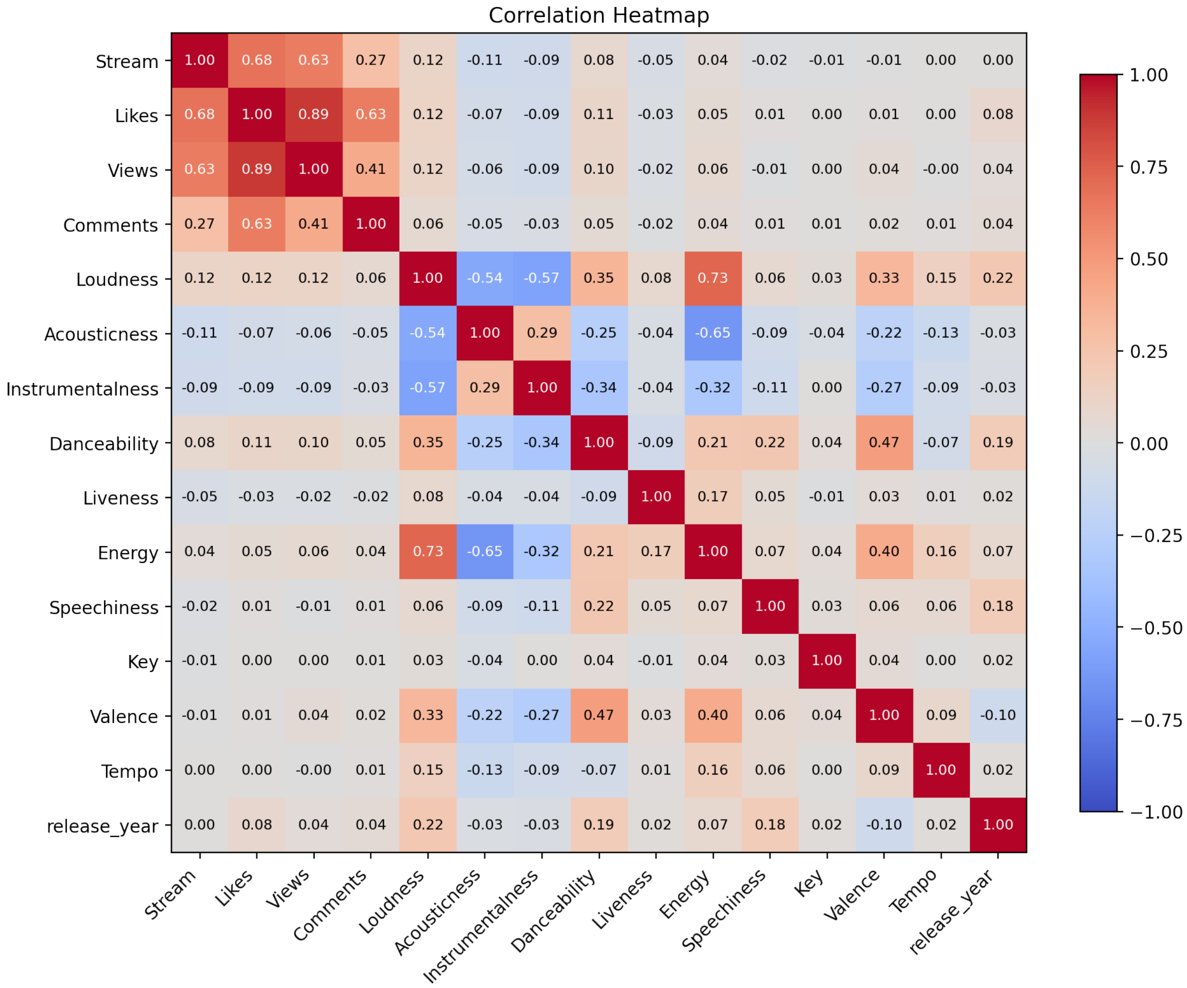

Figure 1: Heatmap showing correlations between platform metrics and audio features.

We create the heatmap to quantify how platform metrics (Streams, Views, Likes, Comments) relate to audio features. The three popularity metrics are tightly clustered: Likes and Views are almost collinear, and both are strongly correlated with Streams. Comments are still related but noticeably weaker, suggesting they reflect a different dimension of engagement. On the audio side, Loudness and Energy are strongly positively correlated, and both are negatively correlated with Acousticness; Instrumentalness also tends to drop as Loudness, Energy, and Danceability rise. Danceability and Valence have a modest positive link. Direct links from audio to engagement exist but are weak overall — features like Danceability, Energy, and Loudness show slight positive correlations with popularity, while Acousticness and Instrumentalness lean slightly negative; Speechiness, Key, Tempo, and Liveness are basically neutral. That implies audio features alone will not strongly predict success without nonlinear effects or interactions.

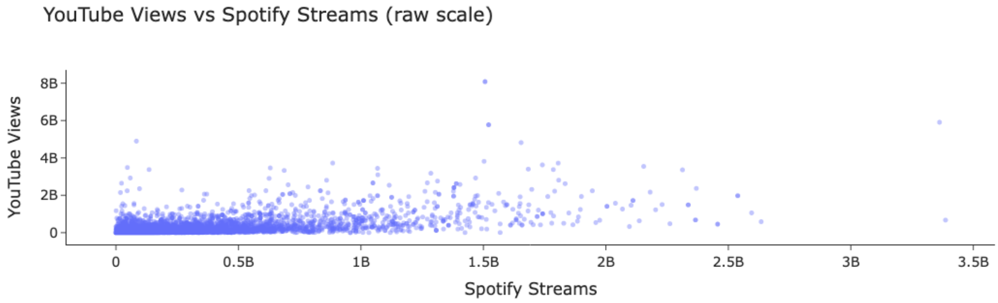

Figure 2: Scatterplot of Spotify streams vs YouTube views.

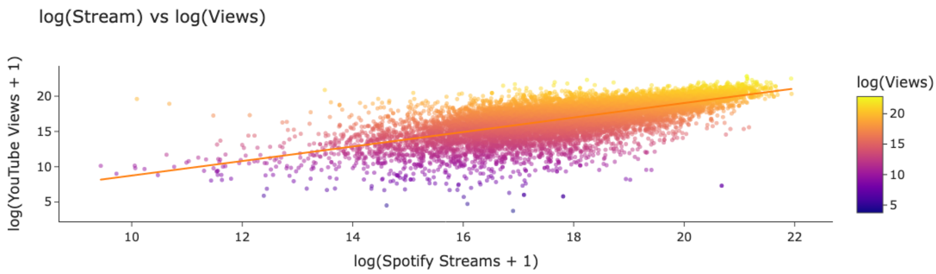

Figure 3: Scatterplot of log-transformed Spotify streams vs YouTube views.

After log scaling both platforms, there is a clear positive relationship between Spotify streaming data and YouTube views, meaning that songs which perform well on Spotify tend to perform well on YouTube. There is considerable variation along this line, with many songs receiving vastly more or fewer views on YouTube than one would expect from merely streaming on Spotify.

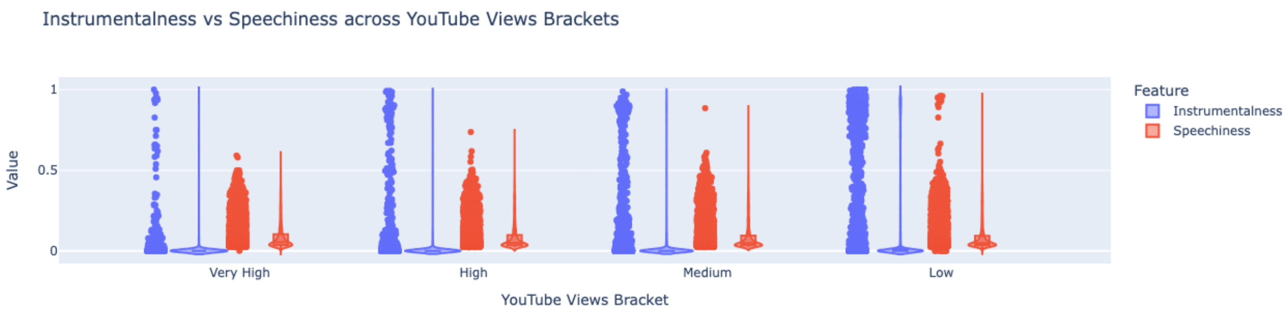

Figure 4: Violin and boxplot overlays of Instrumentalness, Speechiness, Loudness, and Valence by YouTube view brackets

We create a violin and boxplot overlay by YouTube Views Bracket to compare Instrumentalness and Speechiness across popularity levels. Instrumentalness is tightly clustered near zero in every bracket, meaning most tracks are vocal rather than fully instrumental;

purely instrumental tracks do appear but are rare at all popularity levels. Speechiness, by contrast, is lower and less variable in the Very High / High view groups, and it increases in the Low views group.

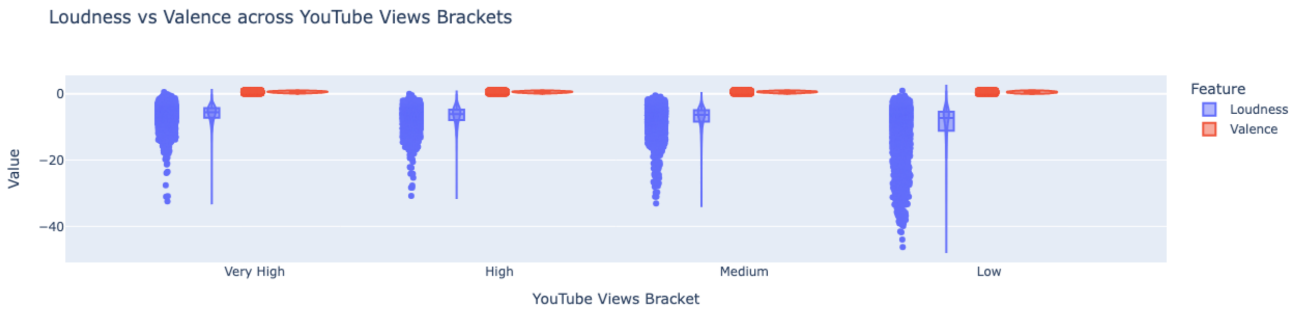

Figure 5: Distribution of Loudness and Valence by YouTube view group

Loudness is tightly packed in the Very High and High brackets, but becomes much more spread out and quieter in Medium and especially Low, where we also see long negative tails that likely reflect low-quality or live recordings. Valence changes less, but its median is slightly higher for the most-viewed brackets and shifts downward in the Low bracket, suggesting that darker songs are more common among lower-view videos.

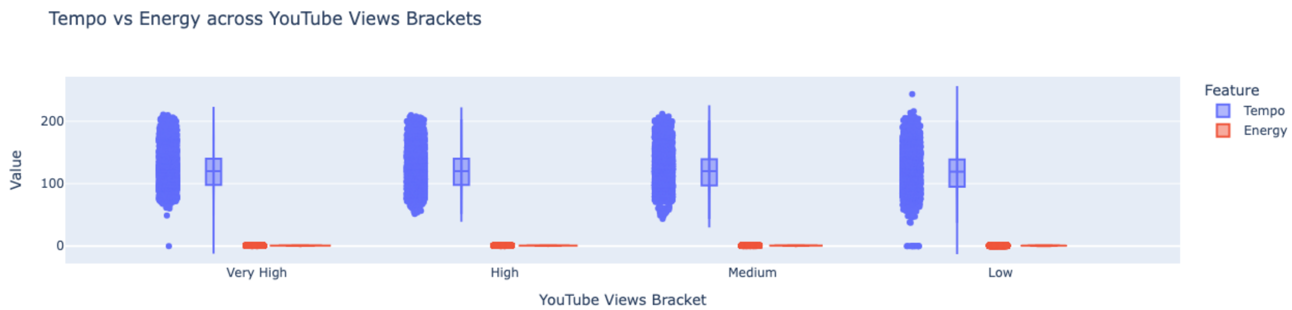

Figure 6: Distribution of Loudness and Valence by YouTube view group

Tempo is all over the place in every bracket — popular songs cover a wide BPM range, and there's no clear trend that faster or slower songs get more views. Energy, though, does separate the groups: Very High and High view brackets have higher median energy and tighter spread, while Medium and Low drop in energy and get noisier. In short, high-performing videos tend to feel more energetic, but not necessarily faster.

Throughout these pairwise views, Energy and Loudness exhibit the most consistent distinctions between best- and worst-performing brackets, and Instrumentalness is nearly everywhere close to zero and non-informative.

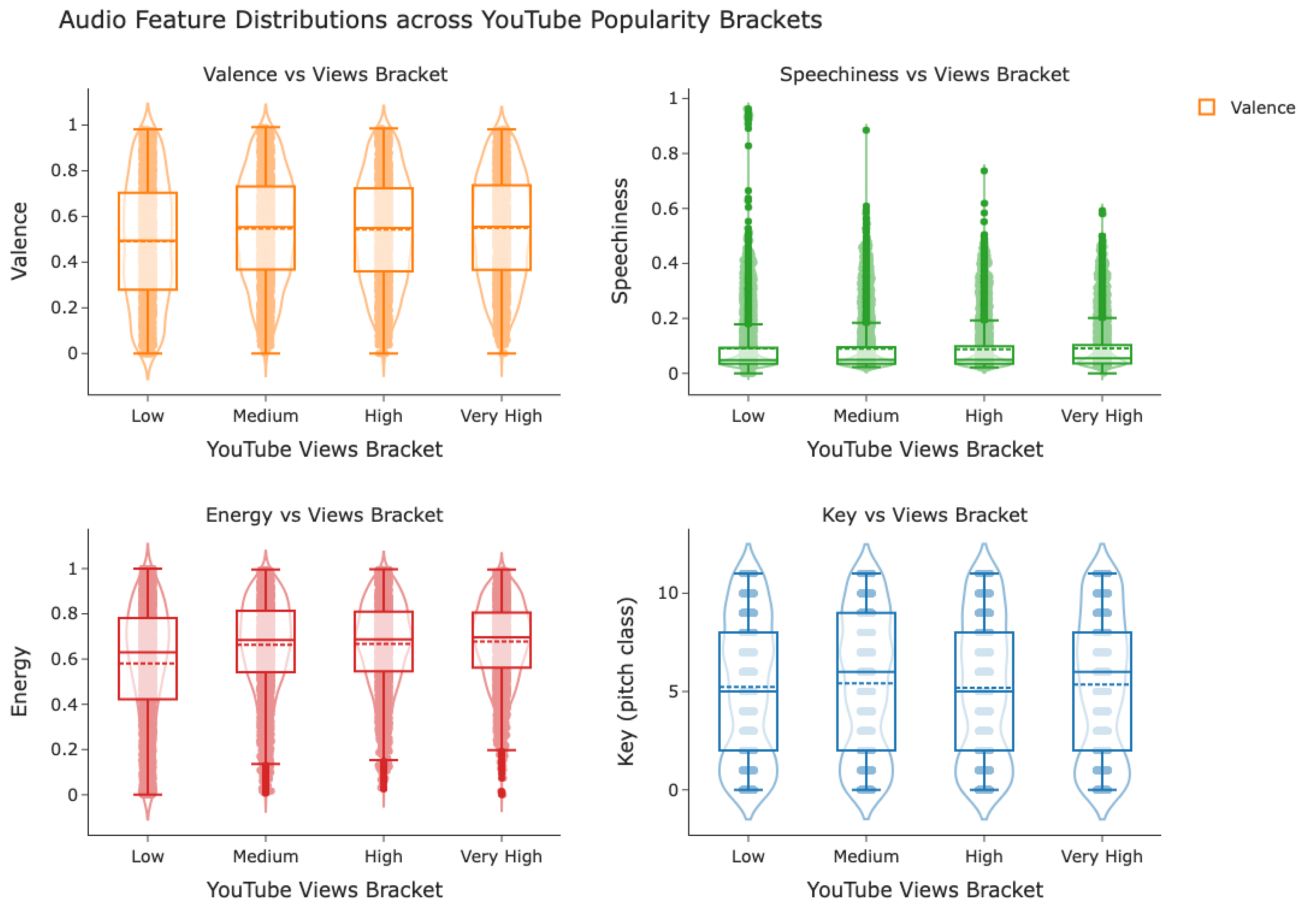

Figure 7: Distribution of key audio features by YouTube view group.

Speechiness increases as a function of views declining, and Valence moves downward slightly, pointing toward mainstream, highly viewed songs as being sung, lively, bright productions that are professional in their loudness. For all YouTube views groups, most audio features cluster closely together. The valence values and musical key contours of low-view and highly viewed tracks are almost indistinguishable. So there is not a clear role for “happier” tracks or specific keys. Energy levels are a touch higher in the highly viewed group, so popular tracks tend to be ever so slightly more energetic. Speechiness levels dip slightly in the most highly viewed group, so highly speechy tracks tend to be less common among the most popular YouTube hits.

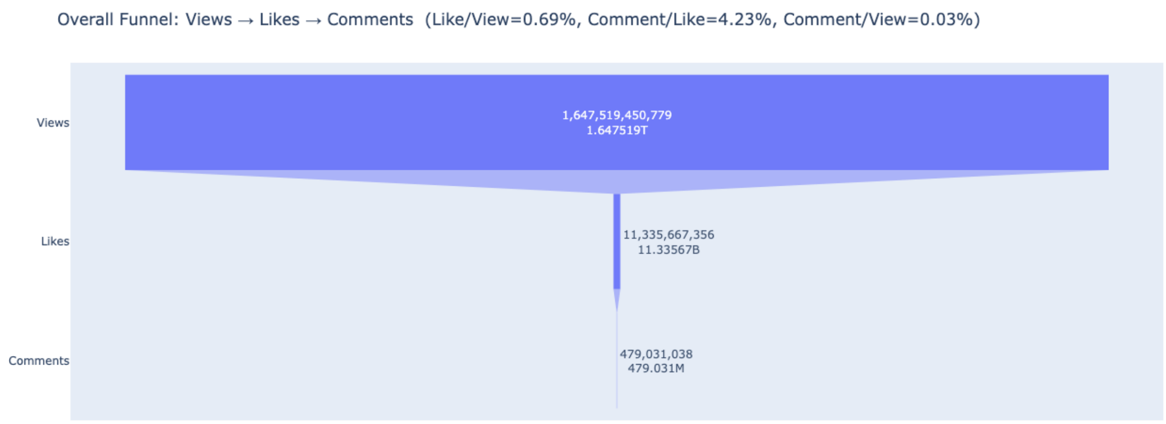

Figure 8: Engagement funnel: conversion from views to likes to comments.

We plot an overall engagement funnel: Views → Likes → Comments. Across the catalog, ∼1.65T views lead to ∼11.3B likes and ∼0.48B comments. That means only ∼0.69% of views turn into likes, and ∼0.03% of views turn into comments. However, once someone has liked, the drop to commenting is less extreme: about 4.23% of likes become comments. In other words, mass exposure rarely turns into active interaction, but people who bother to like are relatively more willing to also comment.

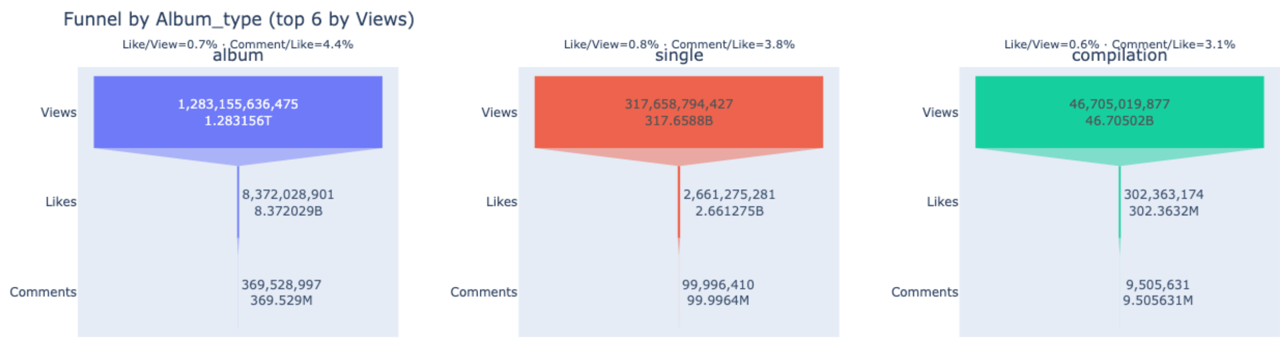

Figure 9: Engagement funnel by release type

We break the engagement funnel out by release format. Albums drive most of the total volume, but their conversion rates are only middle-of-the-road: roughly 0.65–0.7% of views become likes and ∼0.029% of views become comments. Singles have far fewer total views but are the most efficient at turning reach into interaction: ∼0.8%+ of views become likes and ∼0.03% of views become comments. Compilations sit at the bottom, with weaker like and comment rates per view, suggesting more passive consumption.

Together, the findings suggest that extremely low baseline conversion from viewing to interaction is the norm in this context, and that format selection impacts engagement return. Singles have the highest tendency to engender interaction per unit of reach, albums facilitate scale with average conversion, and compilations underperform on a per-view basis. For projects that value active engagement over brute reach, focusing on single launches and reinforcing prompts to like or comment are likely to provide the greatest marginal returns.

4. Modeling Strategies

We assess how Spotify streaming intensity and songs' features (Danceability, Energy, Key, Loudness, Speechiness, Acousticness, Instrumentalness, Liveness, Valence, Tempo, and Duration) are associated with YouTube view counts. Since both Views and Streams are highly skewed and span several orders of magnitude, we model their logarithms. Specifically, we use log(Views) as the outcome and include log(Streams) as a key predictor.

4.1 Fit the Full Model

To fit the full model, we first use a multiple linear regression specification.

Full Model

$$\begin{aligned}\log(\text{Views}) = &-0.5000 + 1.1151\,\text{Danceability} + 0.1359\,\text{Energy} + 0.0036\,\text{Key} + 0.1023\,\text{Loudness}\\ &- 0.1699\,\text{Speechiness} + 0.4126\,\text{Acousticness} - 1.3709\,\text{Instrumentalness} + 0.2530\,\text{Liveness}\\ &+ 0.0005\,\text{Valence} + 0.0015\,\text{Tempo} + 1.046\times10^{-6}\,\text{Duration\_ms} + 0.9402\ \log(\text{Streams}).\end{aligned}$$

	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.5000	0.273	-1.830	0.067	-1.036	0.036
Danceability	1.1151	0.125	8.896	0.000	0.869	1.361
Energy	0.1359	0.142	0.959	0.338	-0.142	0.414
Key	0.0036	0.005	0.790	0.430	-0.005	0.012
Loudness	0.1023	0.007	15.650	0.000	0.089	0.115
Speechiness	-0.1699	0.180	-0.945	0.344	-0.522	0.182
Acousticness	0.4126	0.079	5.194	0.000	0.257	0.568
Instrumentalness	-1.3709	0.111	-12.339	0.000	-1.589	-1.153
Liveness	0.2530	0.101	2.515	0.012	0.056	0.450
Valence	0.0005	0.083	0.006	0.995	-0.162	0.163
Tempo	0.0015	0.001	2.666	0.008	0.000	0.003
Duration_ms	1.046e-06	1.16e-07	9.002	0.000	8.18e-07	1.27e-06
log_Stream	0.9402	0.011	89.019	0.000	0.919	0.961

Figure 10: Screenshot of full model regression summary.

At the 5% significance level, Valence, Speechiness, Energy, and Key are not statistically significant. We therefore fail to reject the null hypothesis that their partial coefficients equal zero, given the other covariates. This indicates insufficient evidence of association with log(Views) in this model. A reduced, more parsimonious model excluding these variables will be compared to the full model using AIC and R^2 to ensure no material loss in fit or generalization.

Cross-Validation

The model achieves $R^2 = 0.569$ on held-out data, indicating that it explains about 56.9% of the variance in log(Views) out of sample.

4.2 Why Choose a Linear Regression Model

We next examine whether standard linear model assumptions are reasonably satisfied.

Linearity and Homoskedasticity

The residual-versus-fitted plot shows the LOWESS curve essentially flat around zero, indicating no systematic curvature and confirming adequacy of the linear functional form. Apparent clustering in the residual cloud is driven by a few high-leverage observations; upon closer inspection, the pattern is diffuse, suggesting no structured heteroskedasticity beyond outliers.

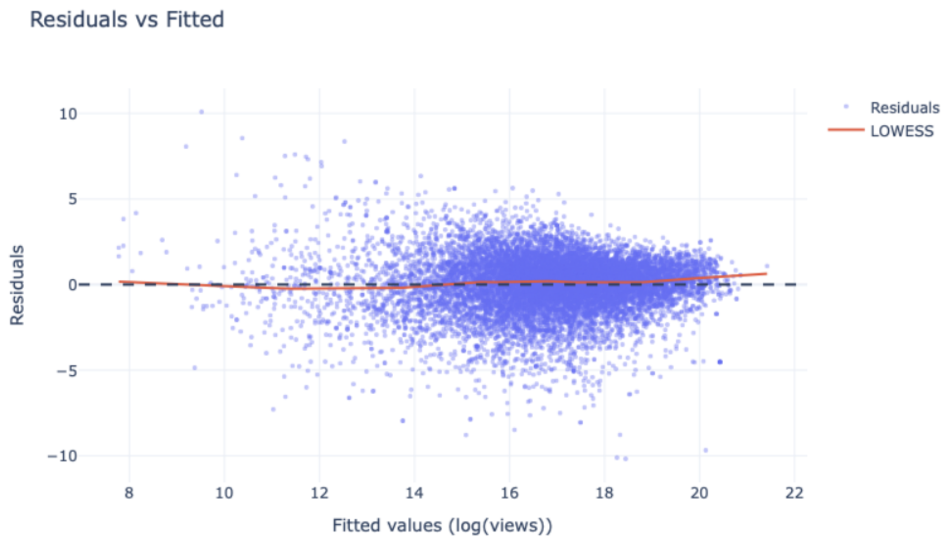

Figure 11: Residual vs Fitted Value Plot of Full Model.

Independence / No Autocorrelation of Errors

We apply the Box–Ljung test to assess independence of residuals. The null hypothesis is that residuals are independently distributed. The p-value is approximately zero, smaller than 0.05, leading us to reject the null and conclude that residuals are not independently distributed.

Table 1: Ljung–Box statistics (lags h)

h	lb_stat	lb_pvalue
5	2725.139866	0.0
10	2842.890741	0.0
20	2846.210960	0.0

Breusch–Godfrey p-value: **0.0**

Multicollinearity

Variance Inflation Factors (VIF) indicate substantial multicollinearity, especially for log(Streams) (57.5), Energy (25.5), Danceability (21.5), and Tempo (17.2). We thus adopt a more parsimonious specification, trimming redundant features while verifying that the key log(Streams) coefficient remains stable and model fit unchanged under cross-validation.

Table 2: Variance Inflation Factors (VIF)

idx	feature	VIF
0	log_Stream	57.503452
1	Energy	25.489561
2	Danceability	21.451806
3	Tempo	17.157685
4	Loudness	9.001412
5	Valence	8.922577
6	Duration_ms	3.496518
7	Acousticness	3.254265
8	Key	3.194128
9	Liveness	2.452395
10	Speechiness	2.093587
11	Instrumentalness	1.748368

4.3 Model Selection via AIC

We next use the Akaike Information Criterion (AIC) to select the preferred model:

$$\begin{aligned}\log(\widehat{\text{Views}}) = &-0.2188 + 0.1097\,\text{Loudness} + 0.9491\,\log(\text{Streams})\\ &- 1.2953\,\text{Instrumentalness} + 1.0392\,\text{Danceability}\\ &+ 1.022 \times 10^{-6}\,\text{Duration\_ms} + 0.3225\,\text{Acousticness}\,.\end{aligned}$$

	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.2188	0.179	-1.219	0.223	-0.571	0.133
Loudness	0.1097	0.005	24.176	0.000	0.101	0.119
log_Stream	0.9491	0.009	107.485	0.000	0.932	0.966
Instrumentalness	-1.2953	0.097	-13.383	0.000	-1.485	-1.106
Danceability	1.0392	0.095	10.910	0.000	0.853	1.226
Duration_ms	1.022e-06	1.01e-07	10.150	0.000	8.25e-07	1.22e-06
Acousticness	0.3225	0.061	5.247	0.000	0.202	0.443

Figure 12: Summary of AIC Model.

All retained coefficients have p-values less than 0.05, so we reject the null for each. The reduced model's AIC is 50,591.78, while the full model's AIC is 39,790; since lower AIC indicates better trade-off between fit and complexity, the full model remains preferred.

4.4 Outlier Removal and Retesting

Outliers were then removed, and the model was refitted:

$$\begin{aligned}\log(\widehat{\text{Views}}) = 0.8079 &+ 0.9066 \log(\text{Streams}) + 0.0402\,\text{Loudness}\\ &- 0.6494\,\text{Instrumentalness} + 0.5686\,\text{Danceability}\\ &+ 6.273 \times 10^{-7}\,\text{Duration\_ms} + 0.5516\,\text{Acousticness}\,.\end{aligned}$$

	coef	std err	t	P>|t|	[0.025	0.975]
const	0.8079	0.154	5.235	0.000	0.505	1.110
log_Stream	0.9066	0.008	120.870	0.000	0.892	0.921
Loudness	0.0402	0.004	9.022	0.000	0.031	0.049
Instrumentalness	-0.6494	0.101	-6.442	0.000	-0.847	-0.452
Danceability	0.5686	0.082	6.912	0.000	0.407	0.730
Duration_ms	6.273e-07	8.23e-08	7.624	0.000	4.66e-07	7.89e-07
Acousticness	0.5516	0.054	10.214	0.000	0.446	0.657

Figure 13: Summary of AIC Model after outlier removal.

The model's AIC is approximately 28,590, lower than earlier models, indicating an improved fit-complexity balance.

4.5 Assumption Checks for AIC Model (After Outlier Removal)

Linearity and Homoskedasticity

The residual-versus-fitted plot again shows the LOWESS curve flat around zero, indicating no systematic curvature. Minor clustering remains but without structured patterns, demonstrating that the revised model mitigates earlier dependence issues.

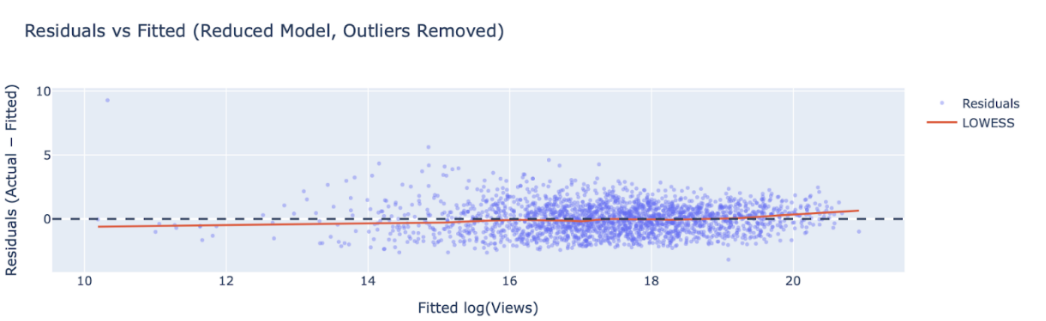

Figure 14: Residual vs Fitted Value after outlier removal.

Independence / No Autocorrelation of Errors

After removing outliers, the Box–Ljung tests report: $Q(5) = 0.832$, $Q(10) = 0.937$, $Q(20) = 0.976$. We therefore do not reject the null of no autocorrelation.

Table 3: Ljung–Box statistics (lags h)

h	lb_stat	lb_pvalue
5	2.119991	0.832305
10	4.207574	0.937497
20	9.541770	0.975741

Multicollinearity

Multicollinearity is no longer a concern. All VIFs range from approximately 1.0 to 1.44 (maximum: Loudness = 1.44), well below typical thresholds (VIF $\geq$ 5 or 10). Thus, no variable trimming or PCA is needed.

Table 4: Variance Inflation Factors (VIF)

idx	feature	VIF
0	Loudness	1.438200
1	Acousticness	1.260567
2	Instrumentalness	1.186062
3	Danceability	1.086567
4	log_Stream	1.029519
5	Duration_ms	1.014031

4.6 Coefficient Interpretations

- **log(Streams):** A one-unit increase in log(Streams) is associated with a 0.9066 increase in log(Views), holding other variables constant.
- **Loudness:** A one-unit increase in Loudness is associated with about a 0.004 average increase in log(Views), all else equal.
- **Instrumentalness:** A one-unit increase in Instrumentalness is associated with approximately a 0.6494 decrease in log(Views).
- **Danceability:** A one-unit increase in Danceability is associated with approximately a 0.5686 increase in log(Views).
- **Duration_ms:** An additional 10 seconds (10,000 ms) of duration corresponds to about a 0.63% increase in Views; 30 seconds corresponds to roughly 1.9%.
- **Acousticness:** A 0.1 increase in Acousticness is associated with approximately a 5.7% increase in Views.

4.7 Summary

After removing outliers, we fit an AIC-selected linear model for log(Views):

$$\begin{aligned}\log(\widehat{\text{Views}}) &= 0.8079 + 0.9066\ \log(\text{Streams}) + 0.0402\,\text{Loudness} \\ &\quad - 0.6494\,\text{Instrumentalness} + 0.5686\,\text{Danceability} \\ &\quad + 6.273 \times 10^{-7}\,\text{Duration\_ms} + 0.5516\,\text{Acousticness}\,.\end{aligned}$$

This final model is stable and suitable for reporting. Coefficients are interpretable and align well with linear regression assumptions.

5. Discussion

Our model investigates whether Spotify audio features and engagement metrics can predict YouTube view counts for corresponding songs. Using multiple linear regression, the model achieved R^2 values between 0.55 and 0.63 across specifications, suggesting that Spotify data offer meaningful insights into cross-platform popularity. This performance demonstrates that the model captures a substantial portion of the variation in YouTube views, reflecting both strong connections and inherent limitations in predictability.

The results consistently highlight log(Streams) as the most influential predictor, carrying the largest and most statistically significant coefficient. This confirms that listening engagement on Spotify serves as a reliable proxy for general song popularity. Songs with higher stream counts benefit from greater exposure, social sharing, and algorithmic recommendation, all of which reinforce visibility on YouTube.

Beyond stream volume, several audio characteristics contribute significantly to predicting YouTube performance. Features such as Loudness, Danceability, and Acousticness show strong positive associations with view counts, suggesting that energetic, rhythmically engaging, and acoustically rich tracks perform better visually. Conversely, Instrumentalness exhibits a pronounced negative coefficient, indicating that instrumental or lyric-free songs attract less repeat visual engagement. These results align with theories of music consumption emphasizing emotional and lyrical connection as primary drivers of audience response.

Applying AIC improved interpretability and robustness. After removing extreme observations, such as unusually viral or underperforming songs, the model's R^2 increased to about 0.63, and cross-validation confirmed consistency. The refinement process—comprising assumption checks, VIF diagnostics, feature selection, and outlier removal—produced a stable six-variable model (Loudness, log(Streams), Instrumentalness, Danceability, Duration, and Acousticness) that balances fit and parsimony.

6. Limitations

While the analysis reveals meaningful patterns, several limitations constrain the interpretation and generalization of results.

Data Scope and Representativeness

The dataset includes only the top ten songs per artist, ensuring adequate engagement levels but introducing selection bias. These tracks tend to have atypically high stream counts, professional production, and significant marketing support. Thus, the model primarily captures patterns among high-performing songs rather than the broader Spotify catalog.

Feature Limitation and Omitted Variables

Predictors are limited to Spotify's audio features—Danceability, Energy, Loudness, and others—that describe sonic properties but omit behavioral or contextual factors such as listener demographics, playlist placement, and promotional activity. The absence of these factors introduces omitted-variable bias and may overstate the role of purely audio-based predictors.

Temporal Inconsistency

Some tracks were released well before the streaming era and were consumed under different market dynamics. These older songs may not follow modern viewing and listening patterns, potentially distorting relationships between Spotify features and YouTube engagement.

Lack of Temporal Modeling

The analysis treats each song as a static observation, without modeling shifts in musical style or audience preference over time. This limits the model's ability to capture longitudinal changes or evolving genre dynamics.

Missing Video-Related Factors

The model does not account for video quality or presentation—such as cinematography, editing, or narrative—which strongly influence engagement on YouTube. Excluding these factors reduces explanatory power, since visual creativity often amplifies the audio experience.

7. Conclusion

This study demonstrates that Spotify audio features and stream counts can predict YouTube viewership with reasonable accuracy using an interpretable linear regression framework. The positive and statistically significant relationship between Spotify popularity and YouTube engagement supports the notion of cross-platform synergy in music consumption. Sonic characteristics such as loudness, danceability, and acoustic texture further refine predictive accuracy, revealing that energetic and emotionally engaging tracks tend to perform better visually.

From a practical standpoint, the model provides a quantitative foundation for understanding audience behavior across streaming and video platforms. Producers, labels, and strategists can use these insights to anticipate which songs are more likely to succeed cross-platform, guiding marketing and promotional decisions.

Future research should expand the dataset beyond top-performing songs to include a wider variety of artists and genres, integrate richer contextual and behavioral metadata, and apply nonlinear or machine-learning approaches. These enhancements would better capture complex interactions and improve predictive power, yielding a more comprehensive understanding of digital music popularity.